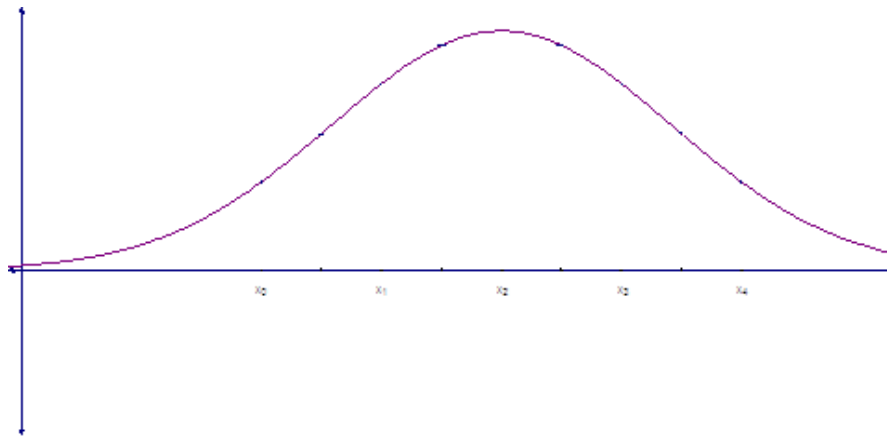


2.6 Numerical Integration: Midpoint, Trapezoid, & Simpson's Rule

Motivation:

In section 2.5 we learned that some indefinite integrals, like $\int e^{-x^2} dx$, cannot be written in terms of elementary functions. So, a good question to ask would be: How can one calculate $\int_a^b e^{-x^2} dx$ to a high level of accuracy?

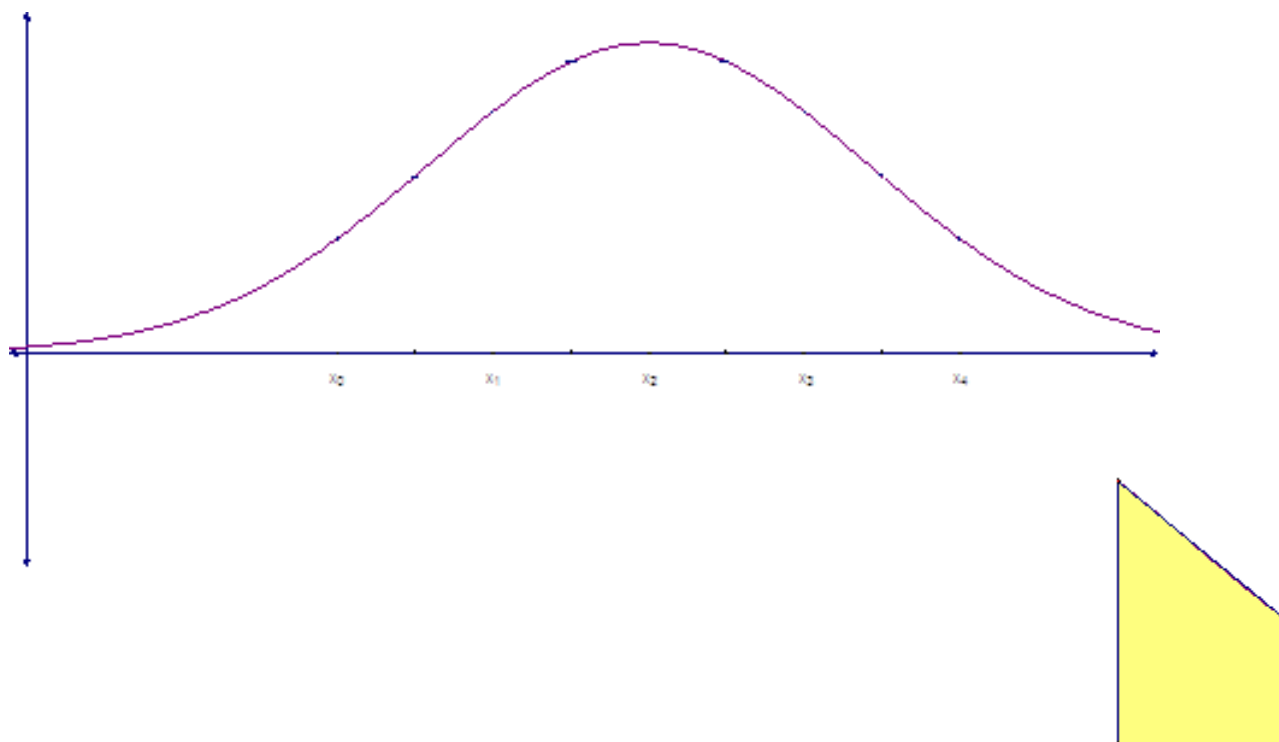
Midpoint Rule



The Midpoint Rule:

Example 1: Use the midpoint rule with $n = 2$ to approximate the value of $\int_{-1}^1 e^{-x^2} dx$.

Trapezoidal Rule



The Trapezoidal Rule:

Example 2:

Use the trapezoidal rule with $n = 2$ to approximate the value of $\int_{-1}^1 e^{-x^2} dx$.

Question to the Class:

Clearly the approximation yielded by the midpoint rule and the trapezoidal rule gets better as the value of n increases. A good question to ask is: “Is there a certain value of n that will guarantee that my approximation is within an error of E from the actual area?”

Theorem: Suppose $|f''(x)| \leq K$ for all x in $[a, b]$. If E_T and E_M are the errors in the Trapezoidal and Midpoint rules, then

$$|E_T| \leq \frac{K(b-a)^3}{12n^2} \quad \text{and} \quad |E_M| \leq \frac{K(b-a)^3}{24n^2}.$$

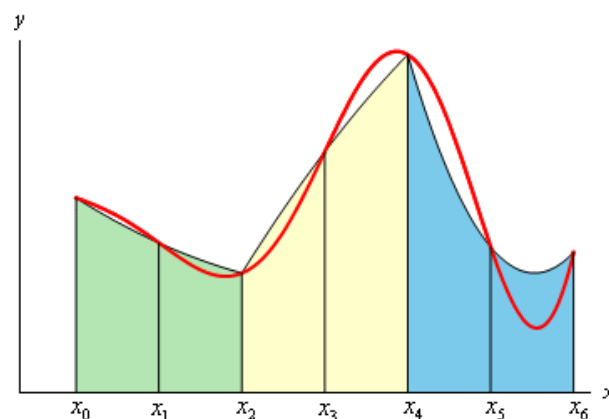
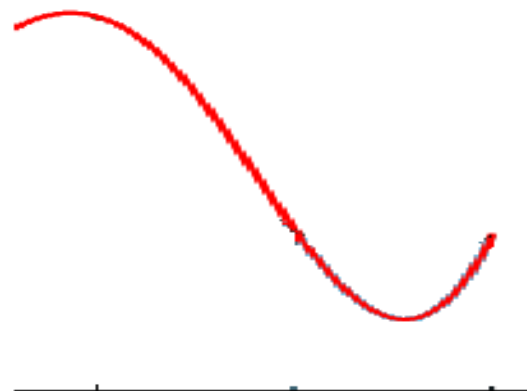
Proof: The proof of this theorem will be withheld.

Example 3:

- (a) Suppose we approximate $\int_{-1}^1 e^{-x^2} dx$ with the midpoint rule and $n = 5$. Give an upper bound for the error involved in this approximation.

- (b) How large should we take n if we want to use the midpoint rule and obtain an approximation of $\int_{-1}^1 e^{-x^2} dx$ that is within $\frac{1}{1000}$ of the actual value of $\int_{-1}^1 e^{-x^2} dx$?

Simpson's Rule



Simpson's Rule: $\int_a^b f(x)dx \approx \frac{\Delta x}{3} (f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \dots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n))$ where n is even and $\Delta x = \frac{b-a}{n}$.

Due to a lack of lecture time we will not derive Simpson's rule. We now present Simpson's rule and the bound on the error (without proof) when Simpson's rule is applied.

Theorem: Suppose $|f^{(4)}(x)| \leq K$ for all x in $[a, b]$. If E_s is the error involved in using Simpson's rule then:

$$|E_s| \leq \frac{K(b-a)^5}{180n^4}.$$

Remark: One should note that the error bound for Simpson's rule has the potential to be *much* smaller than the error bounds for the Trapezoidal and Midpoint rules.

Example 4:

(a) Use Simpson's rule with $n = 4$ to approximate the value of $\int_{-1}^1 e^{-x^2} dx$ (round your answer to four decimal places).

(b) Suppose that you know that if $f(x) = e^{-x^2}$ then $|f^{(4)}(x)| \leq 12$ for any x in $[-1, 1]$. Use this fact to find an upper bound on the error of the approximation you found in (a).